

ECE GY 7363 Communications Networks II: Design and Algorithms

Final Design Project



Objective

- ❑ Apply modeling and solution techniques learned from this course to solve network design problem:
 - routing, dimensioning, topological design, fairness, resilience, restoration multi-layer design, multi-hour design
 - more than just computer networks
 - transportation networks: autonomous car, drones
 - logistic networks
 - online/offline social networks
 -
- ❑ Team Project: 1 -2 students per team, larger team needs permission

Project Format

- ❑ Problem Definition
 - what is the problem?
 - how is it related to this course?
- ❑ Design Models
 - models covered in lectures,
 - other models from textbook
 - variations or new models proposed by you
- ❑ Solution Techniques
 - optimization software: AMPL/CPLEX, Matlab, ...
 - customized heuristics: local search, simulated annealing, genetic algorithms...
 - machine learning models.
 - other numerical solution methods
- ❑ Performance Evaluation
 - instances of problem: network topology, demands, failure situations... (data file in AMPL)
 - performance of your algorithm
 - comparison study, lessons learned.

What to submit?

- ❑ Project Report (all details, no page-limit)
 - problem, model, algorithm, evaluation
- ❑ Project Presentation (10 - 20 slides)
 - highlight of your project
 - in-class presentation (≤ 15 minutes) on May 8th, 9:30am - noon.
- ❑ Code, data files, run script
 - so that I can check your code easily
- ❑ Submit in gradescope
 - one submission each team
 - clearly state contribution of each member
- ❑ Submission Deadline: 11:59pm May 13th, 2026

Evaluation Criteria

- ❑ Novelty & Relevance (20%)
- ❑ Depth & Workload (30%)
- ❑ Correctness (20%)
- ❑ Presentation & Report (30%)

Network Instances for Evaluation

- ❑ Public Datasets Available at
 - <http://sndlib.zib.de/home.action>
- ❑ Many Network Instances
 - small to large network topologies
 - dynamic/multiple traffic matrices for each topology
- ❑ Data can be downloaded in different formats
 - text, XML, and GML
 - java source code

Network Topology

□ other topologies

- e.g. on page 469 of textbook
- <https://www.caida.org/research/topology/#Datasets>
- from research papers, e.g., "Traffic Matrix Estimation: Existing Techniques and New Directions", ACM Sigcomm 2002

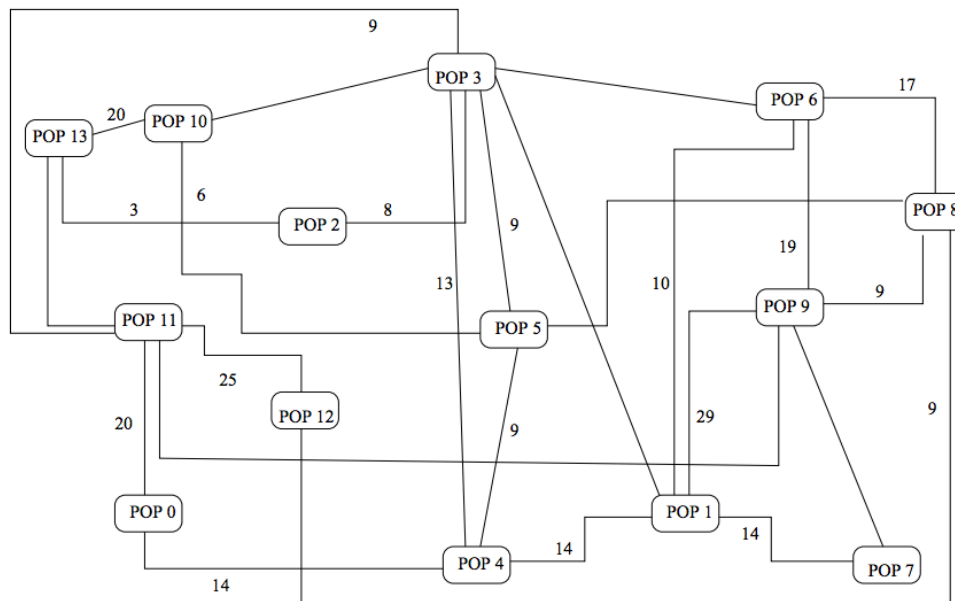


Figure 3: 14-Node Tier-1 POP Topology

Network Topology

- ❑ Generate your own topology
 - random graph, small-world/power law graph,
- ❑ Based on Internet Measurement
 - e.g. <https://www.caida.org/research/topology/>
- ❑ Use topology generator,
 - https://www.cs.bu.edu/brite/user_manual/node3.html
 - BRITE: <https://www.cs.bu.edu/brite/>
 - GT-ITM: <http://www.cc.gatech.edu/fac/Ellen.Zegura/graphs.html>
 - <http://topology.eecs.umich.edu/>

Network Demands

- ❑ Data available on the Internet
 - e.g. <https://www.caida.org/research/traffic-analysis/>
- ❑ Generate your own traffic matrix
 - random, Gaussian, heavy tailed
- ❑ Models from research papers
 - gravity model
 - traffic between A and B is proportional to the product of total traffic of A and total traffic of B
 - "Fast Accurate Computation of Large-Scale IP Traffic Matrices from Link Loads", ACM Sigmetrics 2003

Related Conferences/Workshops

❑ INFOCOM

- <https://infocom2023.ieee-infocom.org/program/accepted-paper-list-main-conference>

❑ SIGCOMM

- <https://conferences.sigcomm.org/sigcomm/2022/program.html>

❑ NSDI

- <https://www.usenix.org/conference/nsdi23/technical-sessions>

❑ ICNP

- <https://icnp22.cs.ucr.edu/acceptedpapers.html>

Related Conferences/Workshops

❑ HOTNETS

- <https://conferences.sigcomm.org/hotnets/2022/accepted.html>

❑ CONEXT

- <https://conferences2.sigcomm.org/co-next/2022/#!/program>

❑ NETWORKING

- <https://networking.ifip.org/2022/index.php/program.html>

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